

Business Problem Statement

A company operating across multiple regions, products, customers, and sales channels generates a large volume of sales transactions. However, management lacks a centralized view of business performance, making it difficult to identify revenue drivers, profitable products, high-value customers, regional trends, and operational inefficiencies.

Without data-driven insights, decision-makers face challenges in:

- Identifying top-performing and underperforming products
- Understanding regional sales variations
- Evaluating channel effectiveness
- Monitoring profitability and profit margins
- Recognizing valuable customer segments
- Forecasting future growth opportunities
- Allocating resources effectively

Deliverables

1. Data Preparation & Modeling (Python): Clean and transform the raw dataset for analysis.
2. Data Analysis (SQL): Organize the data into a structured format, simulate business transactions, and run queries to extract insights on customer segments, loyalty, and purchase drivers.
3. Visualization & Insights (Power BI): Build an interactive dashboard that highlights key patterns and trends, enabling stakeholders to make data-driven decisions.
4. Report and Presentation: Write a clear project report summarizing your key findings and business recommendations. Prepare a presentation that visually communicates insights and actionable recommendations to stakeholders.
5. GitHub Repository: Include all Python scripts, SQL queries, and dashboard files in a well-structured repository